TDMA Schedule Planner and Optimizer for Wireless Networks

Technical Project Report — Wireless Protocol Development Role

Project Report

1. Title & Metadata

TDMA Schedule Planner and Optimizer for Wireless Networks

Internship Assignment — Wireless Protocol Development Role

Implementation Stack: Python 3.11+, NetworkX 3.x, Pytest

Repository Version: 1.0.0

Test Suite: 113 / 113 Tests Passing (`pytest tests/`)

2. Abstract

Time Division Multiple Access (TDMA) is a foundational channel access method in wireless ad-hoc, mesh, and tactical communications. Efficient TDMA scheduling requires assigning time slots to radio nodes such that neither direct interference (Distance-1) nor hidden terminal interference (Distance-2) occurs, while minimizing total frame length to maximize channel capacity.

This project presents a Python 3.11+ application that converts 2-D node positions into a Distance-2 conflict graph and computes a collision-free schedule using the DSATUR graph coloring algorithm coupled with a greedy local search pass. The pipeline outputs structured text reports, JSON representations, and boolean schedule matrices.

For Part 2, an EMANE (Extendable Mobile Ad-hoc Network Emulator) schedule adapter module (`schedule_adapter.py`) was developed to convert optimized slot assignments into official EMANE `<emane-tdma-schedule>` XML events with 1 ms slot timing. Network namespace isolation (`ip netns`) and TAP interface binding designs are documented for Linux deployment.

3. Problem Statement & Functional Requirements

In wireless ad-hoc networks, nodes communicate over a shared RF channel without a central base station. Uncoordinated transmissions lead to two primary types of packet collisions:

1. **Direct Interference (1-hop)**: Nodes within mutual transmission range transmit simultaneously, causing collision at each receiver.

2. **Hidden Terminal Interference (2-hop)**: Nodes out of range of each other transmit simultaneously to a common neighbor, colliding at the intermediate receiver.

Functional Requirements

- **FR-1**: Accept arbitrary 2-D node coordinates from JSON files or strings.

- **FR-2**: Construct a Communication Graph G based on a configurable radio range R_{radio}.

- **FR-3**: Construct a Distance-2 Conflict Graph C encapsulating 1-hop and 2-hop interference.

- **FR-4**: Compute a conflict-free TDMA schedule assigning every node exactly one slot $s \in \{0, 1, \dots, F-1\}$.

- **FR-5**: Enable spatial reuse where non-conflicting nodes share slots.

- **FR-6**: Generate a Boolean Schedule Matrix (Frame Length $\times N$).

- **FR-7**: Dynamically validate that 0 conflict edges are violated.

- **FR-8**: Convert schedule into official EMANE TDMA Schedule XML events with 1 ms slot timing.

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4. Mathematical Formulation

4.1 Communication Graph (G)

Let $V = \{v_1, v_2, \dots, v_N\}$ be a set of N radio nodes with 2-D coordinates $(x_i, y_i) \in \mathbb{R}^2$.

An undirected Communication Graph $G = (V, E_G)$ is defined where an edge $(u, v) \in E_G$ exists if and only if:

$$\text{Euclidean_distance}(u, v) = \sqrt{(x_u - x_v)^2 + (y_u - y_v)^2} \le R_{\text{radio}}$$

4.2 Distance-2 Conflict Graph (C)

The Distance-2 Conflict Graph $C = (V, E_C)$ contains an edge $(u, w) \in E_C$ if and only if the shortest path distance in G satisfies:

$$\text{dist}_G(u, w) \le 2$$

Specifically:

- If $\text{dist}_G(u, w) = 1$, $(u, w) \in E_C$ (Direct Interference).

- If $\text{dist}_G(u, w) = 2$, $\exists v \in V$ such that $(u, v) \in E_G$ and $(v, w) \in E_G$, so $(u, w) \in E_C$ (Hidden Terminal Interference).

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5. Scheduling Algorithm & Optimization

5.1 DSATUR Graph Coloring Heuristic

Scheduling is solved as a vertex coloring problem on C. We implement the DSATUR (Degree of Saturation) heuristic:

1. **Saturation Degree ($\text{sat}(v)$)**: Number of distinct colors assigned to adjacent nodes in C.

2. **Selection Rule**: At each step, select uncolored node $v \in V$ maximizing $\text{sat}(v)$. Break ties using vertex degree $\deg_C(v)$, then lexicographic node order.

3. **Coloring Rule**: Assign v the smallest non-negative integer color (slot) $s \in \{0, 1, 2, \dots\}$ not used by any neighbor in C.

4. **Update**: Update saturation degrees of uncolored neighbors of v in C.

5.2 Greedy Local Search Optimizer

To reduce frame length after DSATUR coloring, a greedy local search pass iterates through nodes in deterministic order. For each node u, it attempts to reassign u to the smallest slot $s' < s_{\text{current}}$ that is conflict-free in C. Passes repeat until no further slot reduction occurs.

5.3 Complexity Analysis

- **Graph Construction**: $O(N^2)$ distance calculations; $O(N \cdot \Delta_G^2)$ for 2-hop path traversal.

- **DSATUR Coloring**: $O(N^2 \log N)$ maintaining saturation priority queue.

- **Local Search Pass**: $O(K \cdot N \cdot \Delta_C)$ where K is pass count and Δ_C is max degree in C.

- **Overall Time Complexity**: $O(N^2 \log N)$, making it efficient for networks with hundreds of nodes.

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6. Part 1 Experimental Results (16-Node Grid Topology)

6.1 Input Topology (`examples/nodes16.json`)

A 4×4 grid of 16 radio nodes spaced 300 meters apart with radio range $R_{\text{radio}} = 500.0$ meters.

6.2 Optimization Summary

- **Nodes Processed**: 16

- **Communication Links (E_G)**: 42

- **Conflict Graph Edges (E_C)**: 90

- **Optimized Frame Length**: 9 unique timeslots

6.3 Node $\to$ Slot Assignments

```
Node_01: Slot 8    Node_05: Slot 0    Node_09: Slot 3    Node_13: Slot 8
Node_02: Slot 4    Node_06: Slot 1    Node_10: Slot 8    Node_14: Slot 4
Node_03: Slot 5    Node_07: Slot 7    Node_11: Slot 6    Node_15: Slot 5
Node_04: Slot 6    Node_08: Slot 2    Node_12: Slot 7    Node_16: Slot 8
```

6.4 Boolean Schedule Matrix

```
Slot     |  01  |  02  |  03  |  04  |  05  |  06  |  07  |  08  |  09  |  10  |  11  |  12  |  13
|  14  |  15  |  16
-----------------------------------------------------------------------------------------------
```

```
------------------
Slot 00  |  0   |  0   |  0   |  0   |  1   |  0   |  0   |  0   |  0   |  0   |  0   |  0   |  0
|  0   |  0   |  0
Slot 01  |  0   |  0   |  0   |  0   |  0   |  1   |  0   |  0   |  0   |  0   |  0   |  0   |  0
|  0   |  0   |  0
Slot 02  |  0   |  0   |  0   |  0   |  0   |  0   |  0   |  1   |  0   |  0   |  0   |  0   |  0
|  0   |  0   |  0
Slot 03  |  0   |  0   |  0   |  0   |  0   |  0   |  0   |  0   |  1   |  0   |  0   |  0   |  0
|  0   |  0   |  0
Slot 04  |  0   |  1   |  0   |  0   |  0   |  0   |  0   |  0   |  0   |  0   |  0   |  0   |  0
|  1   |  0   |  0
Slot 05  |  0   |  0   |  1   |  0   |  0   |  0   |  0   |  0   |  0   |  0   |  0   |  0   |  0
|  0   |  1   |  0
Slot 06  |  0   |  0   |  0   |  1   |  0   |  0   |  0   |  0   |  0   |  0   |  1   |  0   |  0
|  0   |  0   |  0
Slot 07  |  0   |  0   |  0   |  0   |  0   |  0   |  1   |  0   |  0   |  0   |  0   |  1   |  0
|  0   |  0   |  0
Slot 08  |  1   |  0   |  0   |  0   |  0   |  0   |  0   |  0   |  0   |  1   |  0   |  0   |  1
|  0   |  0   |  1
```

6.5 Schedule Validation Outcome

- **Conflict Edges Inspected**: 90

- **Conflicts Detected**: 0

- **Status**: **CONFLICT-FREE**

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7. Part 2 – EMANE Integration & Linux Namespace Architecture

> **Execution Status**: Live EMANE runtime daemon execution and packet delivery were not performed in the current Windows development environment.

7.1 Linux Network Namespace Isolation Architecture

In EMANE, creating TAP devices directly in the host root network namespace allows the Linux kernel to route IP traffic internally via loopback (`lo`), bypassing EMANE.

To force packets to pass through virtual TAP interfaces and EMANE MAC/PHY layers, each node is assigned an isolated **Linux Network Namespace** (`node-1` .. `node-16`).

```
Linux Network Namespace (node-5)        Linux Network Namespace (node-6)
   ■■■
```

```
  <multiframe datarate="1000000" frequency="2350000000" power="0.0">
    <frame index="0">
      <slot index="0" nodes="5">
        <tx/>
      </slot>
      <slot index="4" nodes="2 14">
        <tx/>
      </slot>
      <slot index="8" nodes="1 10 13 16">
        <tx/>
      </slot>
    </frame>
  </multiframe>
</emane-tdma-schedule>
```

- **`slotduration`**: `1000` μs (1 ms slot duration).

- **`nodes` attribute**: Space-separated NEM IDs assigned to transmit in that slot.

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8. Testing Summary

| Test Module | Test Focus | Count | Status |

|---|---|---|---|

| `test_coloring.py` | DSATUR heuristic & color normalization | 16 | **PASSED** |

| `test_conflict_graph.py` | Distance-1 & Distance-2 edge generation | 13 | **PASSED** |

| `test_emane_adapter.py` | EMANE XML schema, NEM mapping, 1 ms slots | 10 | **PASSED** |

| `test_end_to_end.py` | 16-node grid, spatial reuse, edge cases | 33 | **PASSED** |

| `test_topology.py` | Coordinate parsing & communication graph | 29 | **PASSED** |

| `test_validator.py` | Dynamic conflict detection & matrix checks | 12 | **PASSED** |

| **Total Test Suite** | **Full Regression Run (`pytest tests/`)** | **113** | **PASSED (0.29s)** |

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9. Limitations & Future Work

Limitations

1. **Development Host OS**: EMANE requires Linux TUN/TAP devices and network namespaces; live runtime execution was not performed on Windows.

2. **Static Topology**: Schedules are calculated for static node coordinates. Dynamic mobility requires dynamic schedule updates over multicast.

Future Enhancements

1. **Dynamic Schedule Renegotiation**: Broadcast updated `TDMAScheduleEvent` messages when node coordinates change dynamically.

2. **Multi-Channel TDMA**: Extend DSATUR to assign (Slot, Frequency Channel) tuples to increase network throughput.

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10. Conclusion

This project delivers a complete, mathematically rigorous, collision-free TDMA Schedule Planner and Optimizer. Part 1 achieves spatial reuse and 0 conflicts across 113 automated unit/integration tests. Part 2 provides a fully unit-tested EMANE TDMA Schedule Adapter alongside a complete, defensible Linux network namespace deployment specification.